# Unsupervised Machine Learning

## Supervised Vs Unsupervised ML

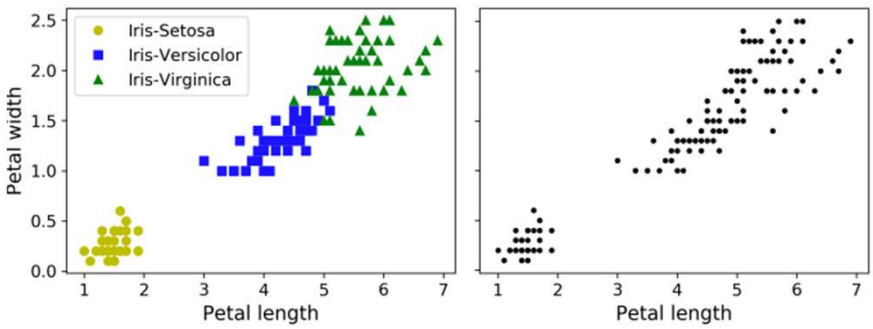
- Supervised ML:
  - Input: feature vectors **X** and classes (labels) **Y**
  - Objective: classify new feature vectors X as belonging to a class of set Y
- But what if the data is not labeled?
- Unsupervised learning:
  - Identify categories in unlabeled data **X**

# Unsupervised ML – Types of Algorithms

- Clustering
  - customer segmentation, recommender systems, search engines, image segmentation, semi-supervised learning, dimensionality reduction, etc.
- Anomaly detection
  - Quality control, outlier detection
- Density estimation

# Clustering

- Identification of similar instances and assigning them to clusters




Labeled data

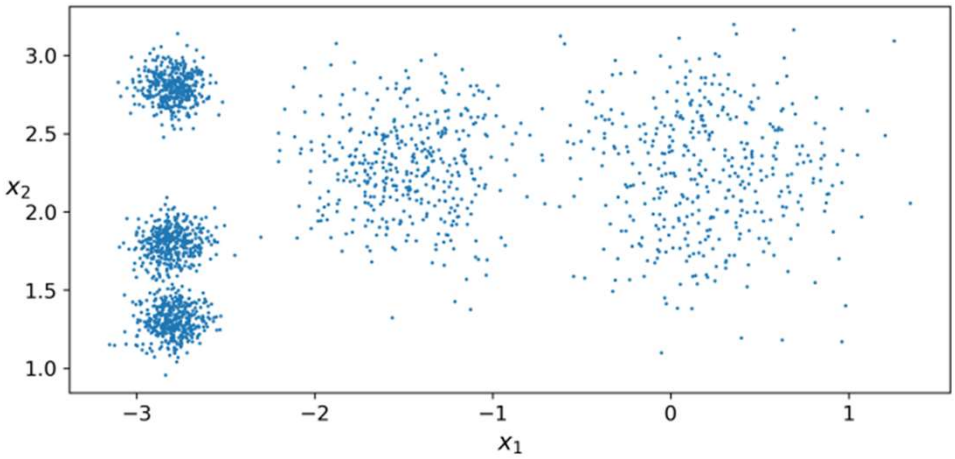
Unlabeled data

# K-Means

- Initialized by randomly placing a number of centroids
  - Centroid == center of a cluster
  - Number of clusters is an input parameter
- Label the data instances as belonging to a particular centroid
- Measure performance
  - Inertia
  - Silhouette score
- If performance is not satisfactory
  - Relocate the centroids and repeat the steps

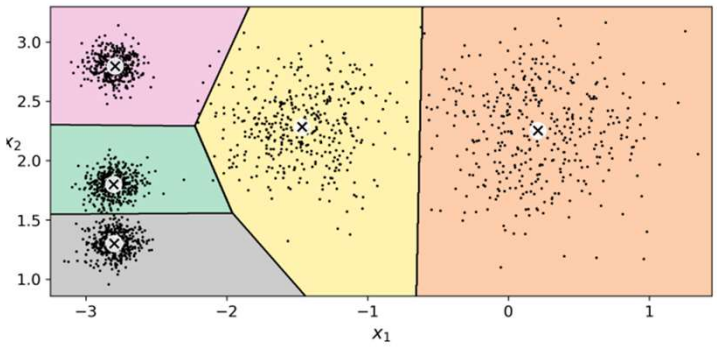
# K-Means

- Test dataset



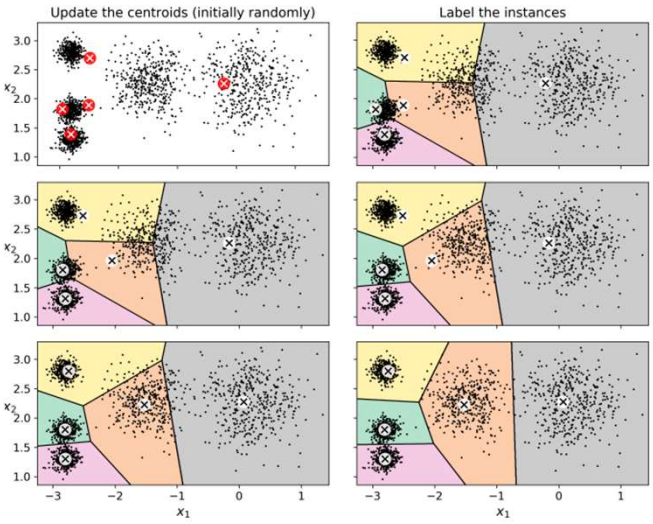
# K-Means

```
from sklearn.cluster import KMeans
k = 5
kmeans = KMeans(n_clusters=k)
y_pred = kmeans.fit_predict(X)
```




```
>>> y_pred
array([4, 0, 1, ..., 2, 1, 0], dtype=int32)
```

# K-Means

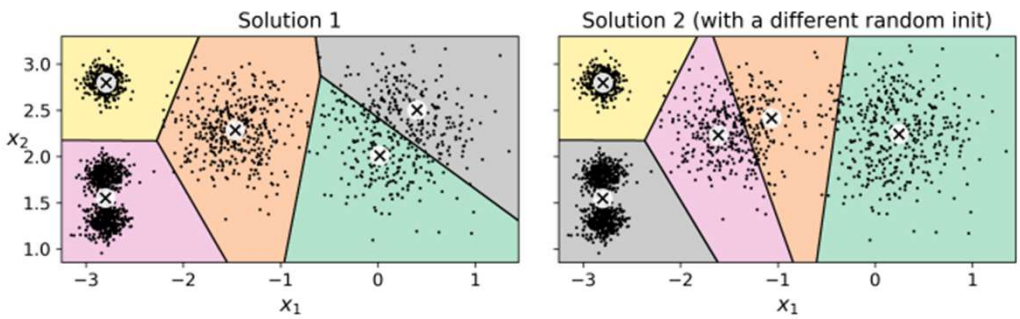


# K-Means

- Complexity
  - Linear (if the data has a clustering structure)
  - Exponential (if the data has little structure)

# K-Means

- Bad centroid initialization



# K-Means – Centroids Initialization

- Manual centroids positioning

```
good_init = np.array([[-3, 3], [-3, 2], [-3, 1], [-1, 2], [0, 2]])
kmeans = KMeans(n_clusters=5, init=good_init, n_init=1)
```

- Try several random positions

# K-Means Inertia

- Mean squared distance between each instance and its centroid.
- The lower the better
- Accessible via a field of the kmeans class

```
>>> kmeans.inertia_
211.59853725816856
```

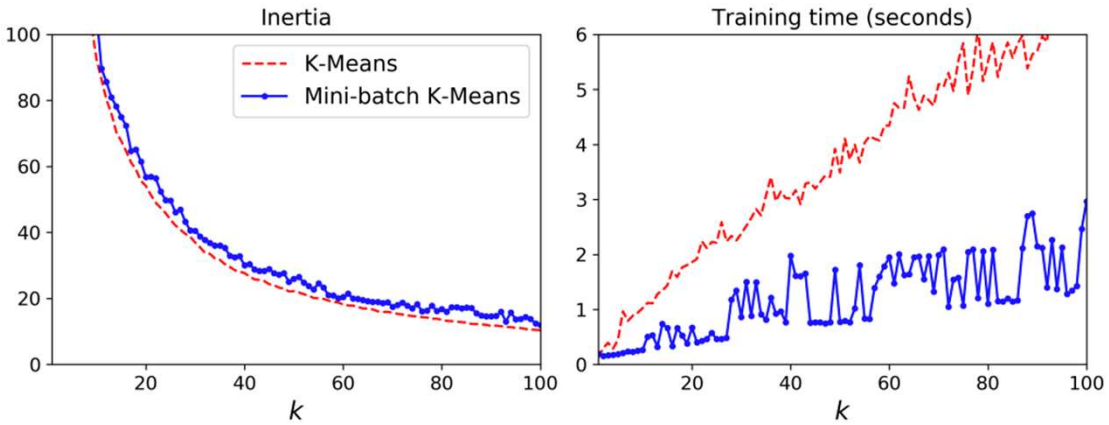
# K-Means+\+

- An improved version of K-Means
- Includes a preliminary step to select centroids that are distant from one another
- Drastically reduces chances of suboptimal convergences
- Is the default initialization method in Scikit-Learn KMeans class

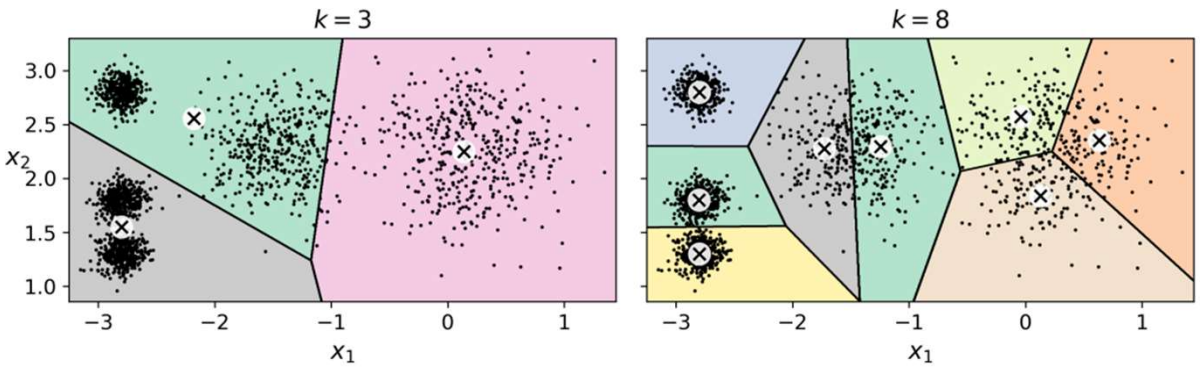
# K-Means – Making It Faster

- Charles Elkan in 2003
  - Speed up distance calculations by the triangle inequality
  - Default in the SciKit Learn KMeans
- David Sculley in 2010
  - Use mini-batches instead of the entire dataset
  - Speed-up factor of 3 to 4
  - Makes possible processing large datasets (tha do not fit in RAM)
  - SciKit Learn implementation: MiniBatchKMeans

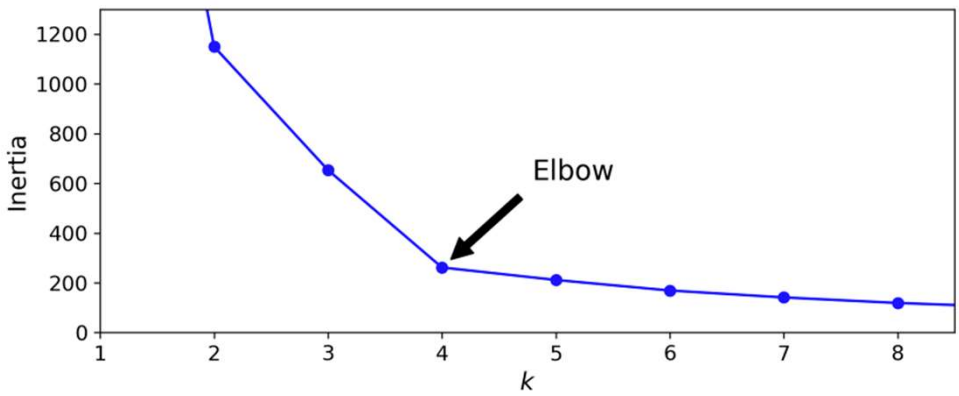
# K-Means vs Mini-Batches K-Means




# K-Means – Optimal Number of Clusters



# K-Mans – Elbow Rule

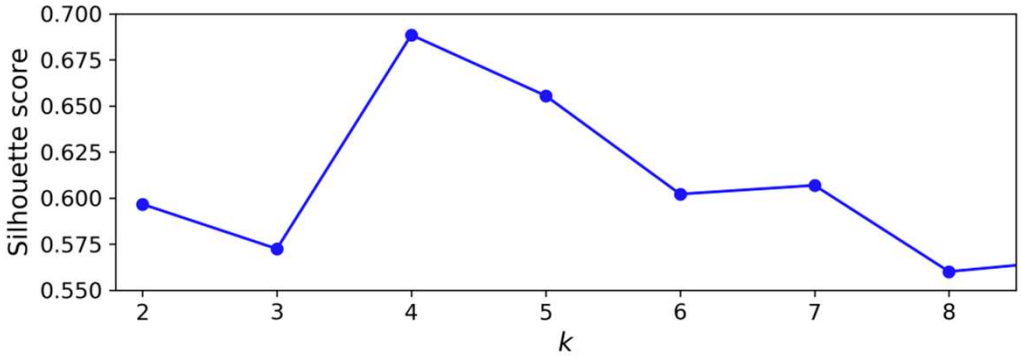



# K-Means - Silhouette Score

- Silhouette Score = mean of silhouette coefficient of all the instances
- Silhouette coefficient
  - For each instance: (b-a)/max(b,a)
  - a = the mean distance to the other instances in the same cluster
  - b = the mean distance to the instances of the next closest cluster

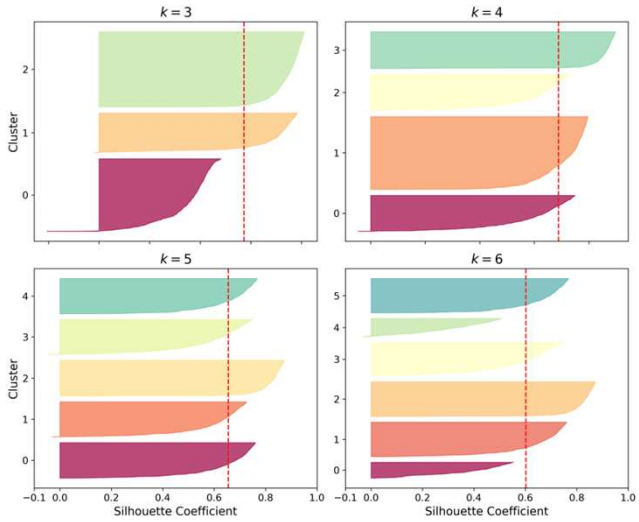
  - Ranges in [-1, +1]
  - +1 == the instance is well inside its own cluster
  - -1 == the instance is likely well outside its own cluster
  - 0 == the instance is close to a cluster boundary

# K-Means - Silhouette Score



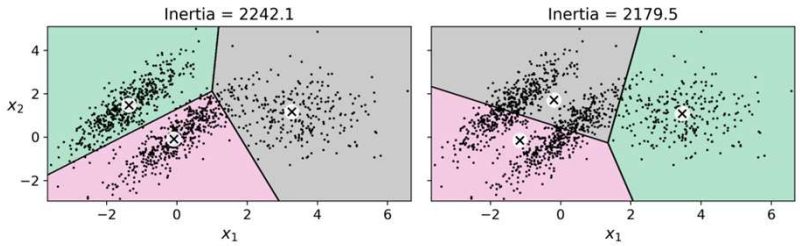

# K-Means - Silhouette Score

Silhouette diagram



# K-Means – Limitations

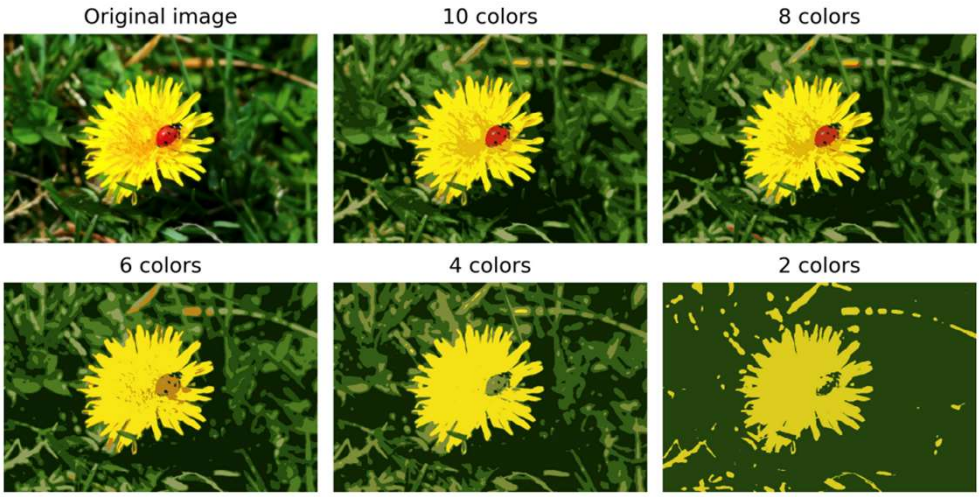
- Selecting the number of centroids is a hassle
- Several runs are required to reach a good solution
- Does not work well with clusters of
  - Varying sizes
  - Different densities
  - Non-spherical shapes




# Clustering In Image Segmentation

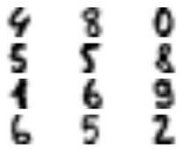
- Partitioning an image into multiple segments
- Semantic segmentation:
  - all pixels that are part of the same object type get assigned to the same segment
- Instance segmentation
  - all pixels that are part of the same individual object are assigned to the same segment
- Color segmentation
  - all pixels of similar color are assigned to same segment

# Clustering In Image Segmentation




# Clustering In Preprocessing

- Example: Handwritten digits classification (MNIST)
  - Digit Image -> distances of the data instance to all centroids



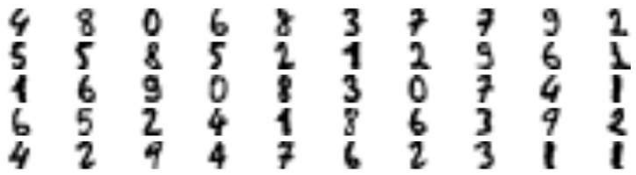
  - Accuracy of logistic regression classifier on raw data:
    - 0.9666
  - Accuracy of logistic regression classifier on K-Means clustered data with 50 centroids
    - 0.9822
  - Accuracy of logistic regression classifier on K-Means clustered data with 90 centroids
    - 0.9844

# Clustering In Semi-Supervised Learning

- Clustering can add labels (centroid indices) to unlabeled data
- Example: 50 samples from handwritten digits (MNIST)
  - Accuracy of logistic regression classifier on raw data:
    - 0.8266
  - Accuracy of logistic regression classifier on only 20% of **representative** instances after K-Means clustered data with 50 centroids
    - 0.9422
  - (Accuracy of logistic regression classifier on the entire dataset with 1797 instances is 0.9666)

# Representative Instances

- Run K-Means with 50 centroids
- Select instances (images) from each cluster that are closest to centroids
- Manually label these images



**Images closests to their centroids**

```
y_representative_digits = np.array([4, 8, 0, 6, 8, 3, ..., 7, 6, 2, 3, 1, 1])
```

## Representative Instances

- Select 20% of instances from each cluster that a closest to centroids and assign them the respective labels (label propagation)
- We achieved very good accuracy with just 50 labeled images (as opposed to 1 797)

## Active Learning

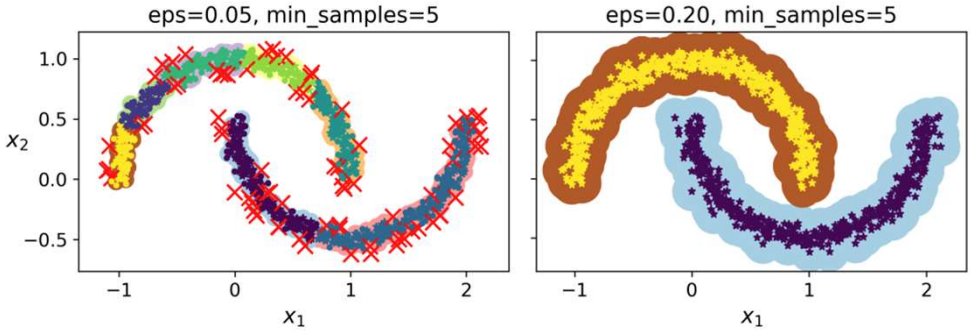
- Machine learning where a human expert interacts with the learning algorithm
- Uncertainty sampling
  - Train the model on labeled instances
  - Apply the model on unlabeled instances
  - Manually label instances for which the model is most uncertain
  - Repeat until satisfactory performance is achieved

# Other Clustering Algorithms

## DBSCAN

- Based on local density extimation
- Can identify clusters of arbitrary shapes
- Hyperparameters
  - eps (epsilon): neighborhood distance
  - min_samples: minimum number of samples for the instance to be considered a **core instance**
- A chain of neighboring core instances makes up a cluster

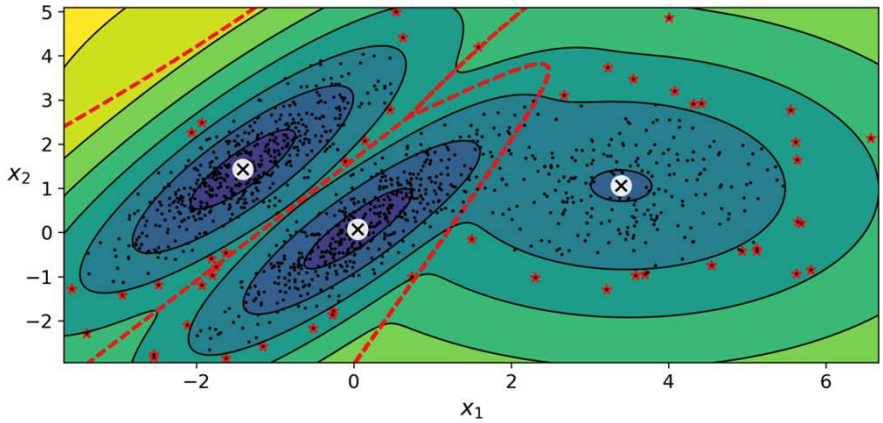
# DBSCAN




- Application area
  - Dense clusters separated by low-density regions
- Scikit-Learn implementation complexity O(m2) for large eps, otherwise O(m log(m))

# Gaussian Mixtures Model

- Each cluster is a Gaussian Distribution



# Gaussian Mixtures Model

- Application area:
  - Datasets of ellipsoid clusters with arbitrary orientation
  - Density estimation
  - Clustering
  - Anomaly detection